

ELECTRONIC DESIGN AND DRAFTING

(EEE437)

COMPILED

BY

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What is design?

The word design stands for a plan of a conceived system or device showing the calculated or selected components that are necessary for it to be created or built

Drafting?

The word drafting stands for the act or discipline of composing diagrams that communicates how a system or device functions or is to be constructed.

Why we study electronic design and drafting: We study electronic design and drafting to be able to design systems and produce standard drafts that can be shared and interpreted by other engineers and technicians globally.

Objectives of this course:

At the end of the course the student should be able to

1. Know the symbols of common components used in electronics.
2. Know how to draw schematic diagrams.
3. Know how to draw block and logic diagrams.
4. Know the construction of a dependable easy to troubleshoot prototype or modification circuit.
5. Know the production of printed circuit boards.
6. Know how to draw wiring assembly diagrams.
7. Understand the various types of design and layout of communication systems.

1.1 Common Electronic Circuit Symbols

In electronic circuits, there are many electronic symbols that are used to represent or identify a basic electronic or electrical device. They are mostly used to draw a circuit diagram and are standardized internationally by the IEEE standard (IEEE Std 315) and the British Standard (BS 3939). NO changes can be brought by the user on any electronic symbol, but the user is free to bring any changes in the architectural drawings like power source and lighting.

	ground		Attenuator		Mosfet		Tube Diode
	Chassis Ground		Capacitor		PNP Bipolar Transistor		Tube Triode
	Battery		Capacitor		NPN Bipolar Transistor		Tube Tetrode
	Resistor		Antenna		P-Channel Junction		Tube Pentode
	Potentiometer		Crystal		N-Channel Junction		Turn-off Diode
	Trimmer		Circuit Breaker		PNP Darlington		SPST
	Fuse		Jack		NPN Darlington		SPDT
	Transducer		Jack		Diode		DPST
	Transducer		Jack		Tunnel Diode		4-Way
	Indicator		Plug		Photodiode/LED		Make Contact
	Bell		Plug		Photosensitive Diode		Break Contact
	Buzzer		Shielded Jack/plug		Inverter		Pushbutton Make
	Fluorescent Lamp		AC Source		Buffer		Pushbutton Break
	Lamp		DC Source		Clock		Relay
	Lamp		Speaker		Function Generator		Reply Coil
	Inductor		Microphone		Summing Amplifier		Transformer
	Half Inductor		Microphone		Logic Gate		
	Pickup Head		Oscillator		Operational Amplifier		
					Operational Amplifier		

Periodic Table of Electronic Symbols

1 Fuse	2 Diode	3 Buffer	4 RS Flip-Flop	5 Resistor	6 Coil, inductor	7 Ferracore coil core	8 Fe-Si coil core	9 Antenna	10 Triode	11 NOT Gate - Inverter	12 JK Flip-Flop	13 Capacitor	14 Variable resistor	15 Adjustable coil	16 Variable coil	17 AC generator	18 Double triode	19 Tri-state buffer gate	20 D Flip-Flop	21 Rectifier diode	22 Zener diode	23 Varicap diode	24 Gunn input diode	25 Schottky diode	26 Constant current diode	27 Instant recovery diode, Snap	28 Tunnel diode	29 Tunnel diode rectifier	30 Schottky diode	31 Trimer	32 Thermistor, NTC	33 Photoresistor, LDR	34 Winding polarity	35 DC voltage generator	36 Octode	37 AND Gate	38 T Flip-Flop	39 Pin diode	40 Photodiode	41 Light Emitting Diode, LED	42 Bicolor dual polarity, LED	43 Laser diode	44 Magnetic diode	45 Temperature sensitive diode	46 Break bidirectional diode PNP	47 Break bidirectional diode NPN	48 Rectifier bridge	49 Variable capacitor	50 Electrolytic capacitor	51 Heating element	52 Nonresistive resistor	53 DC generator	54 Photocell	55 OR Gate	56 DAC digital to analog converter	57 Diac	58 Trigger diac	59 Triac	60 Diacac / Quadac	61 Thyristor SCR, Silicon controlled rectifier	62 Thyristor SCR, Silicon controlled switch	63 Schottky thyristor PNP	64 Inverse conducting thyristor, gate N channel	65 Inverse conducting thyristor, gate P channel	66 Turn-off thyristor gate N channel	67 Turn-off thyristor gate P channel	68 Darlington	69 Photo thyristor	70 SLS, Silicon unilateral switch	71 SLS, Silicon bilateral switch	72 NPN transistor	73 PNP transistor	74 NPN transistor, collector attached cover	75 NPN multi emitter	76 NPN Darlington	77 NPN phototransistor	78 NPN Darlington	79 NPN avalanche transistor	80 NPN Schottky transistor	81 Variable lumped capacitors	82 Polarized capacitors	83 Through capacitor	84 Varistor, VDR	85 Ground / ground line	86 CRT, cathode ray tube	87 CR gate elusive	88 Chip, IC	89 MOSFET, depletion, 3 terminals	90 MOSFET, depletion, 3 terminals	91 MOSFET, depletion, 3 terminals	92 MOSFET, enhancement, coupled with supplier substrate, 3 terminals	93 MOSFET, enhancement, coupled with supplier substrate, 3 terminals	94 MOSFET, depletion coupled substrate supplier, 3 terminals	95 MOSFET, depletion coupled substrate supplier, 3 terminals	96 MOSFET, enhancement, 4 terminals	97 MOSFET, enhancement, 4 terminals	98 MOSFET, depletion, 4 terminals	99 MOSFET, depletion, 4 terminals	100 MOSFET, depletion, 2 gates and 5 terminals	101 MOSFET, enhancement, 2 gates and 5 terminals	102 MOSFET, enhancement, 2 gates and 5 terminals	103 MOSFET, enhancement, 3 terminals	104 LUTP unijunction	105 JFET transistor, channel N	106 JFET transistor, channel N	107 JFET transistor, channel P	108 JFET transistor, channel P	109 JFET transistor, channel P	110 Photo FET	111 NPN tunnel transistor	112 PUT, programmable unijunction
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For more electronic symbols, check www.electronic-symbols.com

Courtesy: Carlos Benes, chesab@gmail.com
References: http://es.wikipedia.org/wiki/Tabela_periódica_de_los_elementos, <http://simbologia-eletronica.com>, <http://www.electronic-symbols.com>



1.2 Electronic Component Identification & Information

Resistors

Standard resistors are NOT polarised and can be placed in the PCB any way around.

The most common have four bands. Bands 1 and 2 identify the first two digits, band 3 is the Multiplier and band 4 is the tolerance (normally Gold 5%)

The Multipliers are:

R = x 0

K = x 1,000

M = x 1,000,000

The kit parts list might contain for example, a 4K7 resistor. The K refers to the number multiplier (K = x 1,000). The position of the K between the 4 and 7 identifies the position of the decimal point so we have

$4.7 \times 1,000 = 4,700$ Ohms (with a colour code of Yellow, Violet, Red)

Another example is say 56R. This gives us $56 \times 0 = 56$ Ohm resistor with colour code Green, Blue, Black.

You will find some five band resistors in our kits and this colour code is likely to be associated with the more precision 1% and 2% types. They work in a similar way to the four band but the first three bands are used to determine the numbers, band 4 is the multiplier and band 5 is the tolerance - see table below for details. Your "garden variety" 5% general purpose types will be four band resistance codes.

Resistor Colour Codes

Five Band Precision Resistor

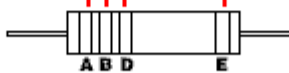
Note: (1) Bands A to D are grouped together.
(2) Band E is tolerance



	Band A	Band B	Band C	Band D	Band E	Band F
Colour	First Digit	Second Digit	Third Digit	Multiplier	Tolerance	Reliability
Black		0	0	1		
Brown	1	1	1	10	$\pm 1\%$	1%
Red	2	2	2	100	$\pm 2\%$	0.1%
Orange	3	3	3	1000	$\pm 3\%$	0.01%
Yellow	4	4	4	10,000	$\pm 4\%$	0.001%
Green	5	5	5	100,000	$\pm 0.5\%$	
Blue	6	6	6	1,000,000	$\pm 0.25\%$	
Violet	7	7	7	10,000,000	$\pm 0.1\%$	
Gray	8	8	8	100,000,000		
White	9	9	9	1,000,000,000		
Gold				0.1	$\pm 5\%$	
Silver				0.01	$\pm 10\%$	
No Colour					$\pm 20\%$	
	Band A	Band B	Band C	Band D	Band E	

Standard Four Band Resistor

Note: (1) Bands 1 to 3 are grouped together.
(2) Band 4 is tolerance



Five Band Resistor with Reliability Band

Note: (1) These are composition type resistors.
(2) Bands are evenly spaced



Resistor Networks

Many resistor networks have an orientation marked with a dot at the end of one side. This dot must be marks pin 1. This pin is marked on the PCB by a square box at one end of the main box.

Capacitors

Capacitor conversion table

Capacitors values may be designated in micro-farad (uF), nano-farad (nF) or pico-farad (pF) and there is a certain amount of overlap. There are many instances where two components of the same value may be quoted in different ways: one may be quoted in terms of pico-Farads and another in terms of nano-Farads. For example, 100 nF is the same as 0.1 micro Farads. The chart below quickly helps show what which values are the same and how many nano-Farads makes a micro-Farad and so forth. It can be used as a quick reference guide for capacitors, or any other electronic component when looking at different items from different manufacturers.

Capacitor conversion table for pico-Farads, nano-Farads, and micro-Farads

micro-Farads (mF)	Nano-Farads (nF)	Pico-Farads (pF)
0.000001	0.001	1
0.00001	0.01	10
0.0001	0.1	100
0.001	1	1000
0.01	10	10000
0.1	100	100000
1	1000	1000000
10	10000	10000000
100	100000	100000000

Using the capacitor conversion table, it is possible to quickly check the relationship between two capacitors with different markings. In this way it is possible to see whether they capacitor values are the same or not.

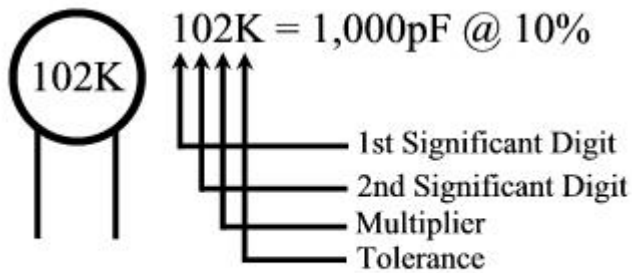
Electrolytic Capacitors

Electrolytic capacitors are POLARISED devices which means they must be inserted with the correct polarity. The long leg denotes the POSITIVE lead and on the opposite side of body is a black band with a minus sign.

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Ceramic Capacitor Codes

How to read a Ceramic capacitor with Numeric coding.



Multiplier Table (Ceramic)

Number	Multiply By (Additional # of Zeros)
0	None (0)
1	10 (1)
2	100 (2)
3	1,000 (3)
4	10,000 (4)
5	100,000 (5)
6	1,000,000 (6)

Common Temperature Coefficient Codes (Ceramic)

Code	Tolerance
C	±0.25pF
J	±5%
K	±10%
M	±20%
D	±0.5pF
Z	+80% / -20%

-

Diodes

All diodes are polarised and must be inserted with the correct orientation to avoid damage to the circuit.

Signal &Y Power Diodes

Diodes are polarised and need to be inserted with the band marked at one end of the body (the cathode) to the band marked on the PCB.

Diodes come all all shapes and sizes but the ones used most often in our kits are 1N4148 signal diodes (orange body with blank band) and 1N4004 rectifier diodes (black with silver band). Both are marked on the body with their part number.

Zener Diodes

You may also find Zener diodes. These look similar to the 1N4148s and are easily muddled up! The voltage is marked on the body e.g. a 5.6V Zener will have 5V6 on the body.

Light Emitting Diodes (LEDs)

Light Emitting Diodes (LEDs) are found in many kits. They are normally red in colour but green and yellow are also sometimes used. LEDs are polarised and must be inserted the correct way.

The negative (cathode) lead is indicated in most cases by a "flat" on that side of the body. This flat is also indicated on the PCB legend for orientation purposes. You may also find that the positive (anode) lead is indicated on the board, corresponding to the long leg of the LED.

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Transistors

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Integrated Circuits

Most Integrated Circuits (ICs) are STATIC SENSITIVE and correct anti-static handling procedures must be observed to avoid damaging them. You might wish to invest in one of the HTS131 Anti-static Wrist Bands that we sell.

Most ICs supplied in our kits are standard DIP Dual Inline Packages.

Unless otherwise instructed ICs should only be inserted once assembly has finished and the board tested (check the voltages present in the IC socket are correct before proceeding). It is important to remove all packing materials including silver foil before inserting the ICs into the IC socket or board. NEVER insert or remove an IC when the board is powered up.

To ensure correct orientation, identify the notch or dot at one end of the body. Pin 1 of the IC is to the left of this marking. Match this to the notch on the board legend and insert the IC, taking care not to bend the legs.

If you need to remove an IC you can use a flat bladed screwdriver. Slide it between the IC socket and the body of the IC and slowly prise it out. Take you time and certainly DO NOT try to pull it out with pliers otherwise you will break legs off.

Miscellaneous

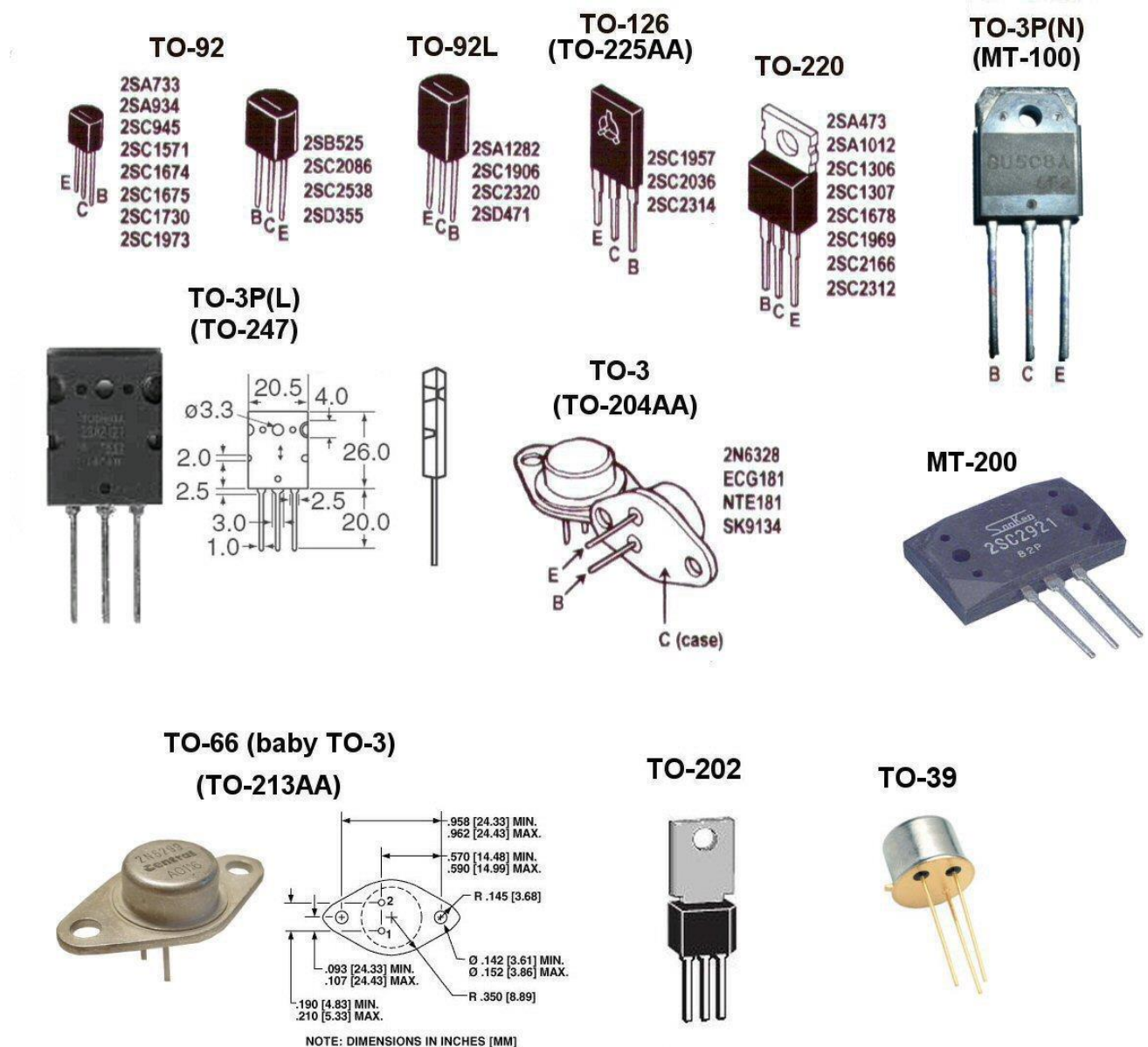
Please see the website http://zeus.cedcc.psu.edu/comp_id_tool.html for help with other components not listed above.

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Common Transistor Case Styles

Note that technically, the case style also indicates the lead layout (CBE, BCE, etc, etc...) Manufacturers flaunt these conventions as it suits them, so do NOT rely on this picture for lead identification. And of course, nothing here is to any kind of scale...

Note about TO-220's. There are a lot of new designs where the traditional metal mounting tab (which is at the same potential as the collector, and in need of electrical isolation in most cases where it contacts a heat sink) is encased in plastic. These plastic-tabbed TO-220's are also called TO-220D's or TO-220F's, and of course do not require electrical insulation of the mounting tab. The only time these cannot be used as a viable substitute for a metal tab TO-220 is in situations where the tab is actually used to conduct current instead of the center collector lead (Sansui does this often on much of their older gear).



Transistor Info.

Right, back to transistors. The three standard transistor marking schemes are:

1. Joint Electron Device Engineering Council (JEDEC).

These take the form:

digit, letter, serial number, [suffix]

The letter is always 'N', and the first digit is one less than the number of legs, (2 for transistors unless they're crippled although I'm not sure about 4 legged transistors maybe they get a 3) except for 4N and 5N which are reserved for optocouplers. The serial number runs from 100 to 9999 and tell nothing about the transistor except its approximate time of introduction.

The (optional) suffix indicates the gain (hfe) group of the device:

A = low gain

B = medium gain

C = high gain

No suffix = ungrouped (any gain).

See the data sheet for the actual gain spread and groupings. The reason for gain grouping is that the low gain devices are fractionally cheaper than the high gain devices, resulting in savings for high volume users.

Examples- 2N3819, 2N2221A, 2N904.

2. Japanese Industrial Standard (JIS).

These take the form:

digit, two letters, serial number, [suffix]

Again, the digit is one less than the number of legs.

The letters indicate the application area and flavor of the device according to the following code:

SA:	PNP HF transistor	SB:	PNP AF transistor
SC:	NPN HF transistor	SD:	NPN AF transistor
SE:	Diodes	SF:	Thyristors
SG:	Gunn devices	SH:	UJT
SJ:	P-channel FET/MOSFET	SK:	N-channel FET/MOSFET
SM:	Triac	SQ:	LED
SR:	Rectifier	SS:	Signal diodes
ST:	Avalanche diodes	SV:	Varicaps
SZ:	Zener diodes		

The serial number runs from 10-9999. The (optional) suffix indicates that the type is approved for use by various Japanese organizations.

NOTE: since the code for transistors always begins with 2S, it is sometimes (more often than not is seems) omitted so, for example, a 2SC733 would be marked C 733.

Examples- 2SA1187, 2SB646, 2SC733.

3. Pro-electron.

These take the form:

two letters, [letter], serial number, [suffix]

The first letter indicates the material:

A = Ge

B = Si

C = GaAs

R = compound materials.

Needless to say the biggest majority of transistors begin with a B.

The second letter indicates the device application:

A:	Diode RF
B:	Variac
C:	transistor, AF, small signal
D:	transistor, AF, power
E:	Tunnel diode
F:	transistor, HF, small signal
K:	Hall effect device
L:	Transistor, HF, power
N:	Optocoupler
P:	Radiation sensitive device
Q:	Radiation producing device
R:	Thyristor, Low power
T:	Thyristor, Power
U:	Transistor, power, switching
Y:	Rectifier
Z:	Zener, or voltage regulator diode

The third letter indicates that the device is intended for industrial or professional rather than commercial applications. It is usually a W,X,Y or Z. The serial number runs from 100-9999. The suffix indicates the gain grouping, as for JEDEC.

Examples- BC108A, BAW68, BF239, BFY51.

Apart from JEDEC, JIS and Pro-electron, manufacturers often introduce their own types, for commercial reasons (ie to get their name into the code) or to emphasize that the range belongs to a specialist application.

Common brand specific prefixes are:

MJ: Motorola power, metal case

MJE: Motorola power, plastic case

MPS: Motorola low power, plastic case

MRF: Motorola HF, VHF and microwave transistor

RCA: RCA

RCS: RCS

TIP: Texas Instruments power transistor (plastic case)

TIPL: TI planar power transistor

TIS: TI small signal transistor (plastic case)

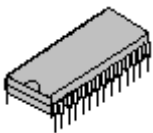
ZT: Ferranti

ZTX: Ferranti

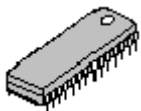
Examples- ZTX302, TIP31A, MJE3055, TIS43.

Many manufacturers also make custom parts for large volume OEM use. These parts are optimized for use in a given part of a given circuit. They usually just have a manufacturers stamp and an untraceable number. Often when a company goes bankrupt, or has surplus at the end of a production run, these transistors find their way into hobbyist bargain packs. There is no way that you can trace data on these devices, so they are only suitable as LED drivers, buffers, etc, where the actual parameters are not important. Check carefully before buying.

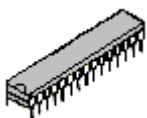
Integrated Circuit Packaging



DIP Dual Inline Package



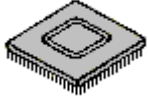
SH-DIP Shrink DIP



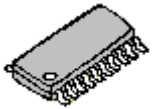
SK-DIP, SL-DIP Skinny DIP, Slim DIP



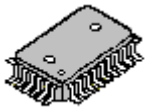
SIP Single Inline Package



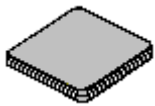
PGA Pin Grid Array



SOIC, SOP Small Outline Package



QFP Quad Flat Package



LCC Leadless Chip Carrier



PLCC, SOJ Plastic Leaded Chip Carrier, Small Outline J-lead Package



IC Common Part Number Codes

Many part numbers for ICs contain some common elements.

The examples below are for the 4xxx series of ICs. (i.e. CD74HC4000 or CD4000B)

Prefixes

CD

CMOS Digital

MM

MOS Monolithic

MN

CMOS

74

Industrial Applications (0-70 deg C) 74xx/54xx TTL Series Pin Compatible

HC

High Speed CMOS as fast as LS TTL family (Low Power Schottky)

Suffixes

B

Buffered Output



Manufacturer's Date Code

The manufacturer's Code is usually a four-digit number. This code is the date of the run in which a particular IC was manufactured. The first two digits represent the year, and the last two digits represent the week. For example a code of 9504 would represent the fourth week of 1995 (the last week of January).

1.3 Circuit Construction

Types of Circuit Board

Breadboard-Temporary, no soldering required

This is a way of making a temporary circuit, for testing purposes or to try out an idea. No soldering is required and all the components can be re-used afterwards. It is easy to change connections and replace components. Almost all the Electronics Club projects started life on a breadboard to check that the circuit worked as intended.

Uses of Breadboard

A breadboard is used to make up temporary circuits for testing or to try out an idea. No soldering is required so it is easy to change connections and replace components. Parts will not be damaged so they will be available to re-use afterwards.

Almost all the Electronics Club projects started life on a breadboard to check that the circuit worked as intended.

The photograph shows a typical small breadboard which is suitable for beginners building simple circuits with one or two ICs (chips). Larger sizes are available and you may wish to buy one of these to start with.



Small Breadboard
Photograph © Rapid Electronics

Connections on Breadboard

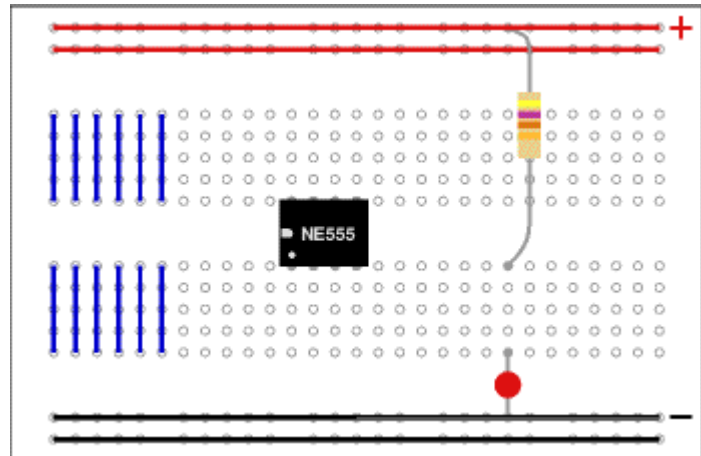
Breadboards have many tiny sockets (called 'holes') arranged on a 0.1" grid. The leads of most components can be pushed straight into the holes. ICs are inserted across the central gap with their notch or dot to the left.

Wire links can be made with single-core plastic-coated wire of 0.6mm diameter (the standard size). Stranded wire is not suitable because it will crumple when pushed into a hole and it may damage the board if strands break off.

The diagram shows how the breadboard holes are connected:

The top and bottom rows are linked horizontally all the way across as shown by the red and black lines on the diagram. The power supply is connected to these rows, + at the top and 0V (zero volts) at the bottom.

I suggest using the upper row of the bottom pair for 0V, then you can use the lower row for the negative supply with circuits requiring a dual supply (e.g. +9V, 0V, -9V).



The other holes are linked vertically in blocks of 5 with no link across the centre as shown by the blue lines on the diagram. Notices how there are separate blocks of connections to each pin of ICs.

Large Breadboards

On larger breadboards there may be a break halfway along the top and bottom power supply rows. It is a good idea to link across the gap before you start to build a circuit, otherwise you may forget and part of your circuit will have no power!

Building a Circuit on Breadboard

Converting a circuit diagram to a breadboard layout is not straightforward because the arrangement of components on breadboard will look quite different from the circuit diagram.

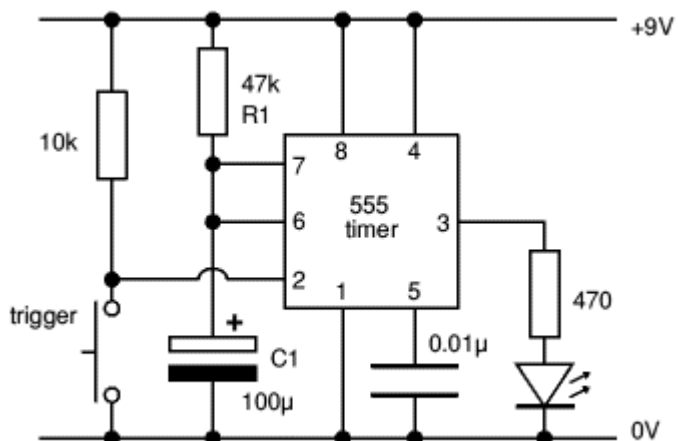
When putting parts on breadboard you must concentrate on their connections, not their positions on the circuit diagram. The IC (chip) is a good starting point so place it in the centre of the breadboard and work round it pin by pin, putting in all the connections and components for each pin in turn.

The best way to explain this is by example, so the process of building this 555 timer circuit on breadboard is listed step-by-step below.

The circuit is a monostable which means it will turn on the LED for about 5 seconds when the 'trigger' button is pressed. The time period is determined by R1 and C1 and you may wish to try changing their values. R1 should be in the range 1kΩ to 1MΩ.

Time Period, $T = 1.1 \times R1 \times C1$

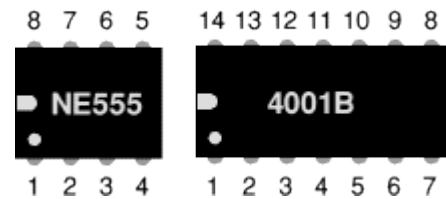
For further information please see 555 monostable.



Monostable Circuit Diagram

IC pin numbers

IC pins are numbered anti-clockwise around the IC starting near the notch or dot. The diagram shows the numbering for 8-pin and 14-pin ICs, but the principle is the same for all sizes.



Components without suitable leads

Some components such as switches and variable resistors do not have suitable leads of their own so you must solder some on yourself. Use single-core plastic-coated wire of 0.6mm diameter (the standard size). Stranded wire is not suitable because it will crumple when pushed into a hole and it may damage the board if strands break off.

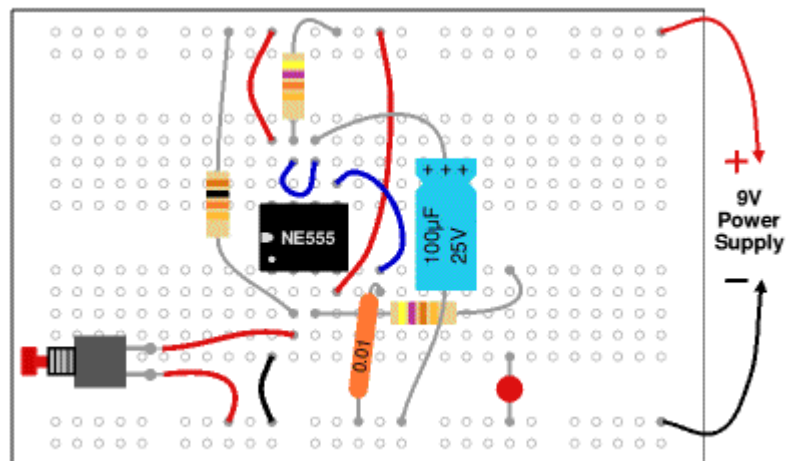


Building the example circuit

Begin by carefully insert the 555 IC in the centre of the breadboard with its notch or dot to the left.

Then deal with each pin of the 555:

1. Connect a wire (black) to 0V.
2. Connect the 10k resistor to +9V. Connect a push switch to 0V (you will need to solder leads onto the switch)
3. Connect the 470 resistor to an used block of 5 holes, then...
Connect an LED (any colour) from that block to 0V (short lead to 0V).



Monostable Circuit on Breadboard

4. Connect a wire (red) to +9V.
5. Connect the 0.01µF capacitor to 0V. You will probably find that its leads are too short to connect directly, so put in a wire link to an unused block of holes and connect to that.
6. Connect the 100µF capacitor to 0V (+ lead to pin 6). Connect a wire (blue) to pin 7.
7. Connect 47k resistor to +9V. Check: there should be a wire already connected to pin 6.

8. Connect a wire (red) to +9V.

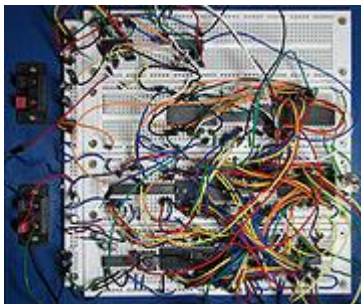
Finally,...

- Check all the connections carefully.
- Check that parts are the correct way round (LED and 100 μ F capacitor).
- Check that no leads are touching (unless they connect to the same block).
- Connect the breadboard to a 9V supply and press the push switch to test the circuit.

If your circuit does not work disconnect (or switch off) the power supply and very carefully re-check every connection against the circuit diagram.

Stripboard

Limitations



An example of a complex circuit built on a breadboard. The circuit is an Intel 8088 single board computer.

Due to large stray capacitance (from 2-25pF per contact point), high inductance of some connections and a relatively high and not very reproducible contact resistance, solderless breadboards are limited to operate at relatively low frequencies, usually less than 10 MHz, depending on the nature of the circuit. The relative high contact resistance can already be a problem for DC and very low frequency circuits. Solderless breadboards are further limited by their voltage and current ratings.

Solderless breadboards usually cannot accommodate Surface mount technology devices (SMD) or non 0.1" (2.54 mm) grid spaced components, like for example those with 2 mm spacing. Further, they can not accommodate components with multiple rows of connectors, if these connectors don't match the DIL layout (impossible to provide correct electrical connectivity). Sometimes small PCB adapters (breakout adapters) can be used to fit the component on. Such adapters carry one or more of the non-fitting components and 0.1" (2.54 mm) connectors in DIL layout. The larger of the components are usually plugged into a socket, where the socket was soldered onto such an adapter. The smaller components (e.g. SMD resistors) are usually directly soldered onto such an adapter. The adapter is then plugged into the breadboard via the 0.1" connectors. However, the need to solder the component



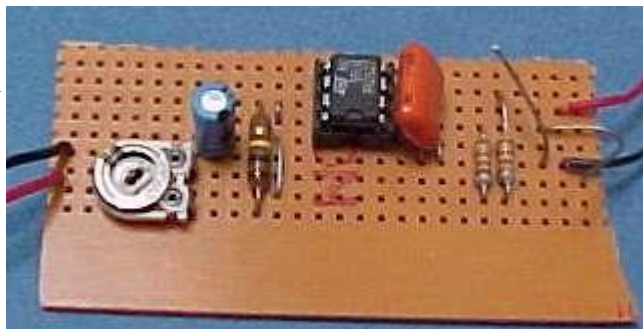
Photograph © Rapid Electronics

or socket onto the adapter contradicts the idea of using a solderless breadboard for prototyping in the first place.

Complex circuits can become unmanageable on a breadboard due to the large amount of wiring necessary.

Stripboard- Permanent, soldered

Stripboard has parallel strips of copper track on one side. The strips are 0.1" (2.54mm) apart and there are holes every 0.1" (2.54mm). Stripboard requires no special preparation other than cutting to size. It can be cut with a junior hacksaw, or simply snap it along the lines of holes by putting it over the edge of a bench or table and pushing hard. Stripboard has parallel strips of copper track on one side. The tracks are 0.1" (2.54mm) apart and there are holes every 0.1" (2.54mm).



Stripboard is used to make up permanent, soldered circuits. It is ideal for small circuits with one or two ICs (chips) but with the large number of holes it is very easy to connect a component in the wrong place. For large, complex circuits it is usually best to use a printed circuit board (PCB) if you can buy or make one.

Stripboard requires no special preparation other than cutting to size. It can be cut with a junior hacksaw, or simply snap it along the lines of holes by putting it over the edge of a bench or table and pushing hard, but take care because this needs a fairly large force and the edges will be rough. You may need to use a large pair of pliers to nibble away any jagged parts.

Avoid handling stripboard that you are not planning to use immediately because sweat from your hands will corrode the copper tracks and this will make soldering difficult. If the copper looks dull, or you can clearly see finger marks, clean the tracks with fine emery paper, a PCB rubber or a dry kitchen scrub before you start soldering.

Placing components on stripboard

Components are placed on the non-copper side, then the stripboard is turned over to solder the component leads to the copper tracks.



Stripboard layouts are shown from the component side, so the tracks are out of sight under the board. Layouts are normally shown with the tracks running horizontally across the diagram.

Placing components on stripboard requires care. The large number of holes means it is very easy to make a mistake! For most small circuits the best method is to very carefully place the IC holder(s) in the correct position and solder in place. Then you can position all the other components relative to the IC holder(s).

Minor position errors left and right will not usually be a problem because the component will still be connected to the correct tracks. However, up and down position errors must be avoided because just one hole too high or too low will connect the component to the wrong track and therefore the wrong part of the circuit.

Some people like to label the holes with letters (up/down) and numbers (across) to give each hole a 'grid reference' but this still requires careful counting of holes.

Planning an example stripboard layout

When planning a stripboard layout, you must concentrate on the connections between components, not their positions on the circuit diagram. The best way to explain the planning process is by example, so the section below explains the process step-by-step for a 555 astable circuit which flashes an LED.

The stripboard tracks are horizontal in all the diagrams.

The circuit diagram

The circuit diagram is the starting point for any stripboard layout, even if you have already built a trial circuit on breadboard.

The LED flashes at a rate determined by the resistors R1 and R2 and the capacitor C1. R1 must be at least 1k Ω and both R1 and R2 should not be more than 1M Ω . To select a value for the LED resistor R3 please see the LEDs page.

LED on time: $T_m = 0.7 \times (R_1 + R_2) \times C_1$

LED off time: $T_s = 0.7 \times R_2 \times C_1$

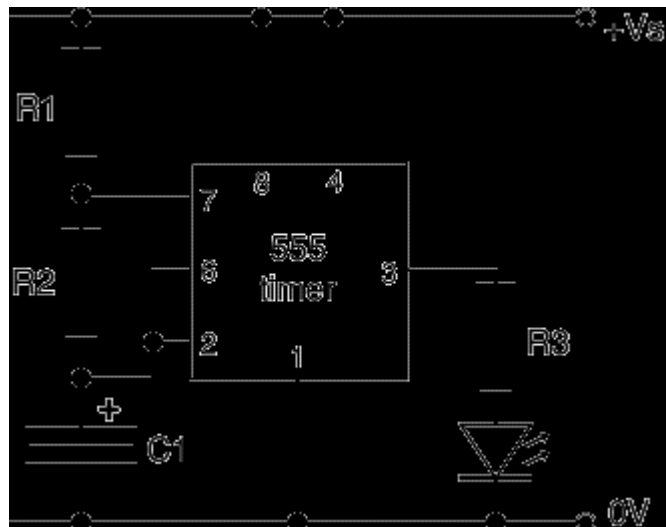
$T = T_m + T_s = 0.7 \times (R_1 + 2R_2) \times C_1$

Frequency (flashes per second), $f = 1/T$

T_m and T_s are about equal if R2 is much larger than R1.

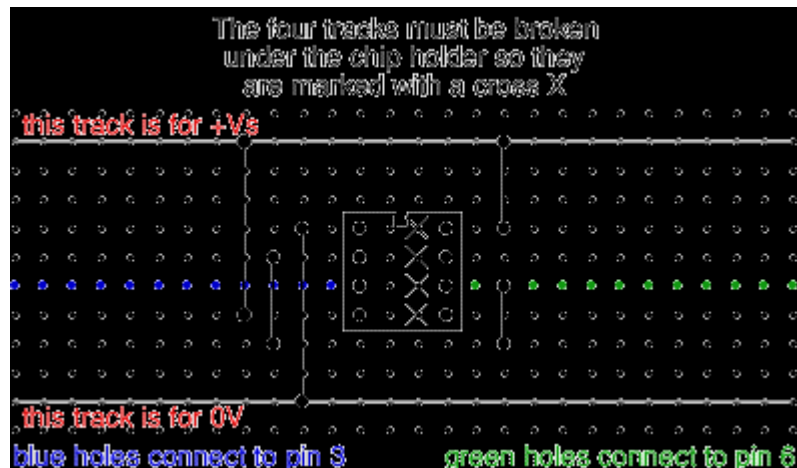
For further information please see 555 astable.

Planning the layout

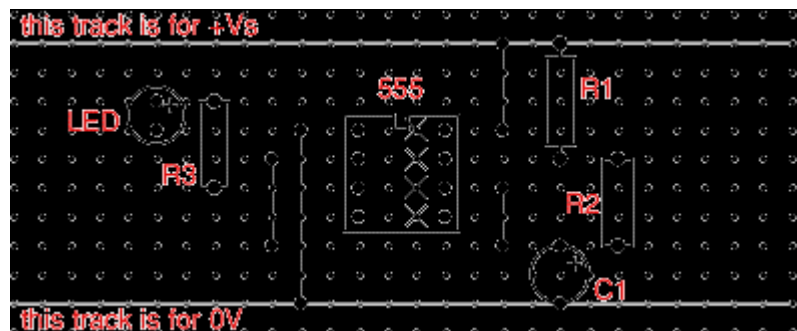


Astable Circuit Diagram

1. Place the IC holder near the centre of your planning sheet with pin 1 at the top left (as in the diagram). You may find it helpful to number the pins.
2. Mark breaks in each track under the IC holder with a cross (X). The breaks prevent opposite pins of the IC being connected together. The track beside each pin of the IC is connected to that pin, the diagram shows this for pins 3 and 6.
3. Mark the power supply tracks +Vs and 0V, choose tracks which are 2 or 3 spaces above and below the IC holder as shown in the diagram.
4. Now add the wire links. Draw a 'blob' (●) at each end of a link. The links are vertical because the stripboard tracks make the horizontal connections. Tinned copper wire (with no insulation) can be used for these links unless there is a risk of them touching other wires (in which case use single core insulated wire). Work round the IC pin-by-pin from pin 1.
 - Draw all the direct links to the supply tracks (+Vs and 0V). The diagram shows pin 1 connected to 0V and pins 4 and 8 connected to +Vs.
 - Draw any links required between pins on the same side of the IC. There are none in the example, but these links are straightforward to add.
 - Links to pins on the other side of the IC require more thought. If the pins happen to be opposite one another you can erase the track break (X) between them. Otherwise the pins can be linked by connecting both of them to an unused track above or below the IC. The diagram shows pins 2 and 6 linked in this way. Another method is to link them with insulated wire bent around the IC (see the Flashing LED project for example).



5. Add components which will be mounted on the stripboard such as resistors, capacitors and diodes. Make sure you allow for their size which determines the minimum number of holes, and sometimes the maximum as well. This is usually the most difficult stage of planning a layout so expect to



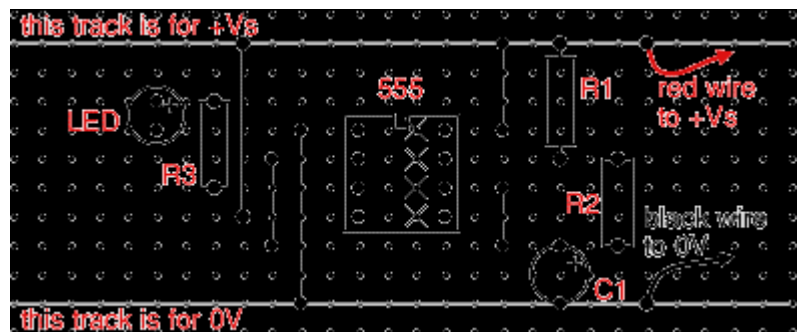
change your plan several times! Remember to label the components, otherwise it will become confusing once there are several on the plan.

Connections which do not involve the IC are made using an unused track. For example resistor R3 and the LED are connected by an unused track above the IC.

Watch for alternative arrangements using the links you have already made. For example the LED needs to connect to 0V but it is a long stretch to the 0V track. It is easier to connect the LED to the same track as pin 1 of the IC because that track is already connected to 0V by a wire link.

Resistor R2 needs to connect from pin 7 to pin 6 and it could do this directly by mounting it vertically. However, it has been connected from pin 7 to the track used to link pins 2 and 6, the extra space this gives allows R2 to lie horizontally on the board.

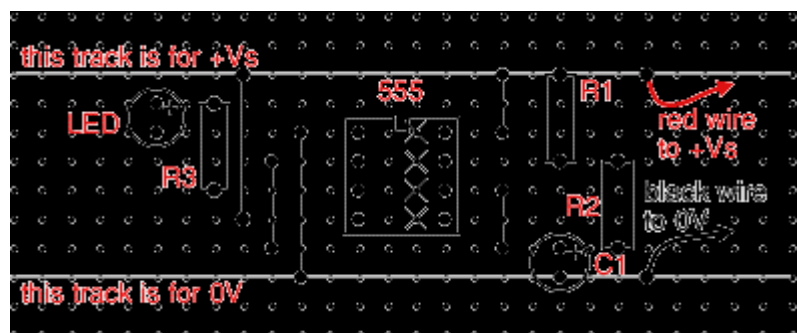
6. Add wires to components which will be off the stripboard such as switches. These should normally be on the left and right at the edges of the board. Start by



adding the battery clip or power supply leads to the +Vs and 0V tracks. Connections for the other off-board components are usually easy because you do not need to allow for their size, just draw wires to the correct tracks.

7. Check your plan very carefully by checking every connection shown on the circuit diagram. A good way to do this is to work round the IC pin-by-pin. Check all the connections and components connected to pin 1, then move on to pin 2, and so on.

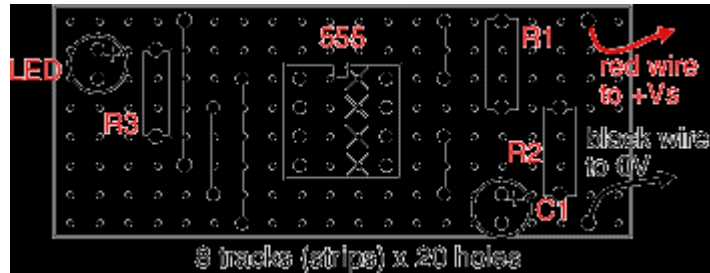
8. Look for ways to improve your plan. For example it may be possible to eliminate an unused track by moving a supply track nearer to the IC - but make sure there is still



sufficient space for the components. It may also be possible to move links and components closer to the IC horizontally to make the area of board required a little smaller.

Unused tracks above and below the IC have been eliminated in the example. This affected two components, resistor R1 and capacitor C1, but both will still fit in the reduced space. The plan could be compressed a little further by moving components and links closer to the IC horizontally but this has not been done.

9. Finally, check your plan again and make a neat copy fully labelled with all the component references or values. Work out the size of stripboard required. Notice that an extra hole has been allowed on the left and right to avoid soldering at the end of a track. Joints made at the end of a track are likely to break because the small piece of track beyond the last hole easily breaks away from the board.

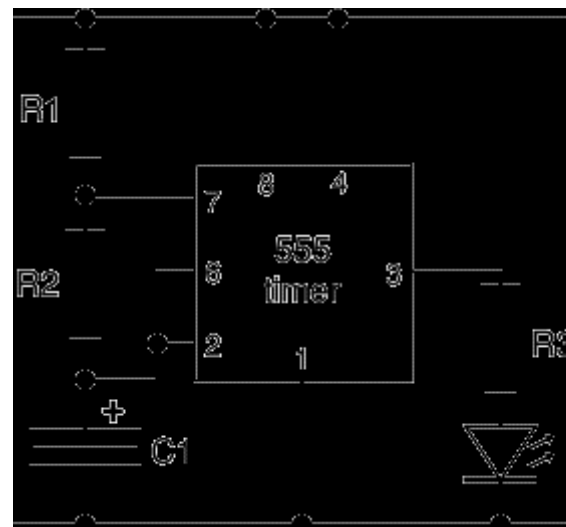
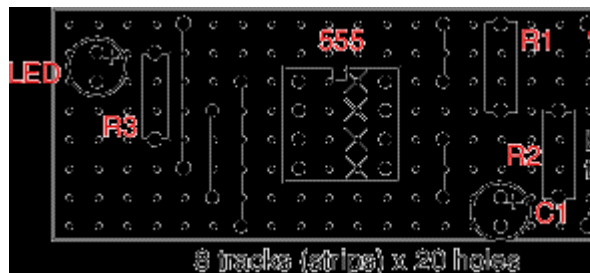


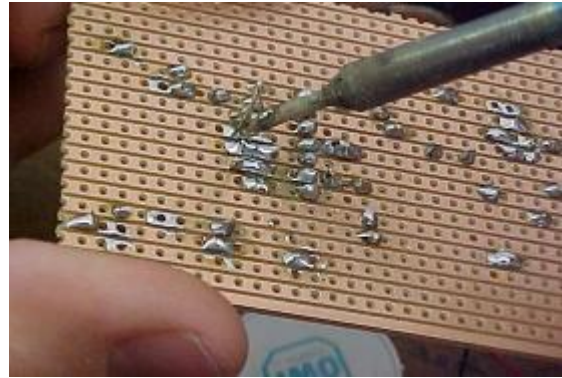
It is tempting to rush straight into soldering the circuit, but do check your plan carefully first. It is much easier to correct errors on the plan than it is to correct to correct them on the soldered board!

This example plan is just one of the many possible layouts for the circuit. The Flashing LED project uses the same circuit, but the stripboard plan is quite different. In this case the aim was to have the minimum number of wire links.

The completed stripboard layout and the circuit diagram for comparison:

Flashing LED Circuit

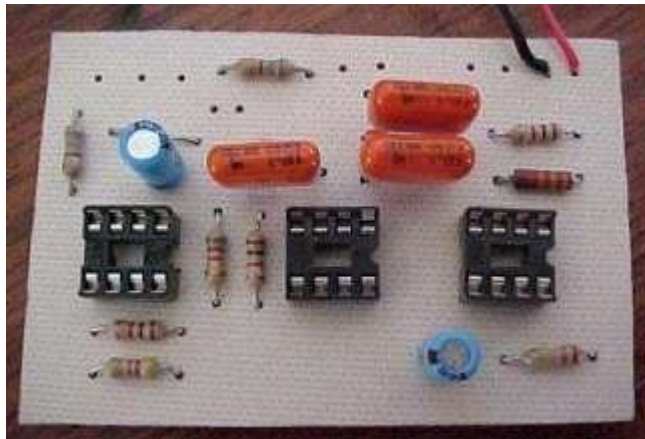




Stripboard circuit (copper tracks side)

Printed Circuit Board-Permanent, soldered

Printed circuit boards have copper tracks connecting the holes where the components are placed. They are designed especially for each circuit and make construction very easy. However, producing the PCB requires special equipment so this method is not recommended if you are a beginner unless the PCB is provided for you.



Design

- Schematic capture or schematic entry is done through an EDA tool.
- Card dimensions and template are decided based on required circuitry and case of the PCB. Determine the fixed components and heat sinks if required.
- Deciding stack layers of the PCB. 4 to 12 layers or more depending on design complexity. Ground plane and Power plane are decided. Signal planes where signals are routed are in top layer as well as internal layers.^[8]
- Line impedance determination using dielectric layer thickness, routing copper thickness and trace-width. Trace separation also taken into account in case of differential signals. Microstrip, stripline or dual stripline can be used to route signals.
- Placement of the components. Thermal considerations and geometry are taken into account. Vias and lands are marked.
- Routing the signal trace. For optimal EMI performance high frequency signals are routed in internal layers between power or ground planes as power plane behaves as ground for AC.
- Gerber File generation for manufacturing.

Patterning (etching)

The vast majority of printed circuit boards are made by bonding a layer of copper over the entire substrate, sometimes on both sides, (creating a "blank PCB") then removing unwanted copper after applying a temporary mask (eg. by etching), leaving only the desired copper traces. A few PCBs are made by *adding* traces to the bare substrate (or a substrate with a very thin layer of copper) usually by a complex process of multiple electroplating steps.

There are three common "subtractive" methods (methods that remove copper) used for the production of printed circuit boards:

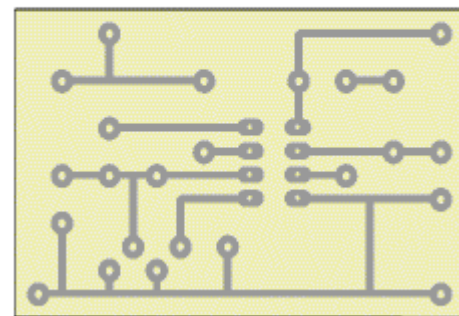
1. Silk screen printing uses etch-resistant inks to protect the copper foil. Subsequent etching removes the unwanted copper. Alternatively, the ink may be conductive, printed on a blank (non-conductive) board. The latter technique is also used in the manufacture of hybrid circuits.
2. Photoengraving uses a photomask and chemical etching to remove the copper foil from the substrate. The photomask is usually prepared with a photoplotter from data produced by a technician using CAM, or computer-aided manufacturing software. Laser-printed transparencies are typically employed for *phototools*; however, direct laser imaging techniques are being employed to replace phototools for high-resolution requirements.
3. PCB milling uses a two or three-axis mechanical milling system to mill away the copper foil from the substrate. A PCB milling machine (referred to as a 'PCB Prototyper') operates in a similar way to a plotter, receiving commands from the host software that control the position of the milling head in the x, y, and (if relevant) z axis. Data to drive the Prototyper is extracted from files generated in PCB design software and stored in HPGL or Gerber file format.

"Additive" processes also exist. The most common is the "semi-additive" process. In this version, the unpatterned board has a thin layer of copper already on it. A reverse mask is then applied. (Unlike a subtractive process mask, this mask exposes those parts of the substrate that will eventually become the traces.) Additional copper is then plated onto the board in the unmasked areas; copper may be plated to any desired weight. Tin-lead or other surface platings are then applied. The mask is stripped away and a brief etching step removes the now-exposed original copper laminate from the board, isolating the individual traces. Some boards with plated thru holes but still single sided were made with a process like this. General Electric made consumer radio sets in the late 1960s using boards like these.

The additive process is commonly used for multi-layer boards as it facilitates the plating-through of the holes (to produce conductive vias) in the circuit board.

Cleaning and Drilling a PCB ready for Soldering

1. Clean off the protective coating from the PCB using a PCB rubber or steel wool so that all the copper tracks are bright and shiny. The PCB rubber has grit in it to make it very abrasive.
In fact the coating can be left on and it should melt away around the joints as you solder, but in the Electronics Club we have generally had better results by removing the coating.



Printed Circuit Board (PCB)

2. Drill the holes with a 1mm diameter bit. This is easiest with a proper electric PCB drill in a stand, but a hand-held miniature electric drill can be used if you take care to avoid twisting and snapping the small drill bit. Wear safety spectacles.
A hand-drill is not suitable for such small bits unless you are very skilled.
3. A few may hole may need to be larger, for example preset resistors usually need a 1.5mm diameter hole. It is simplest to re-drill these special holes afterwards.
4. Check carefully to make sure you find all the holes.
Even with experience it is easy to miss one or two!

2.0 Basic Classification of Electronic Components

The electronic components are mainly classified into two groups or categories. They are
PASSIVECOMPONENTS and ACTIVE COMPONENTS.

ACTIVE COMPONENTS: The components in which transformation of Energy taken place are called active components. Or they can be also defined as, active components are electronic components having the capability of processing a signal its own. Semiconductor devices like BJT, FET, MOSFET and vacuum tube and gas tube devices are listed under Active components.

Eg: Bipolar junction transistor (BJT), FET, Silicon Controlled rectifier (SCR)

PASSIVE COMPONENTS: Components capable of dissipation and storage of energy are called Passive components. Even if active components are capable of processing a signal, they require the assistance of passive components in order to get sufficient voltage levels, energy level, power dissipation etc. at various points.

The proper function, of an active device is decided by the proper values of these passive elements. Hence the selection of these elements such as resistances, inductance, capacitance, and transformers not only require the proper attention, but also decide the proper function of the active devices as well as the circuit as a whole. Eg: Resistors, Capacitors, Inductors

2.1 Schematic diagrams

A drawing showing all significant components (and their interconnections) of a circuit by means of standard symbols.

2.2 Signal flow is the path a signal takes from source to output; the direction of flow is indicated by an arrow.

A subassembly drawing is drawing depicting how the components making up a specific subassembly connect together.

2.3 Guidelines for drawing a schematic diagram

The Schematic of the circuit design shall have the following requirements:

Drawing Rules

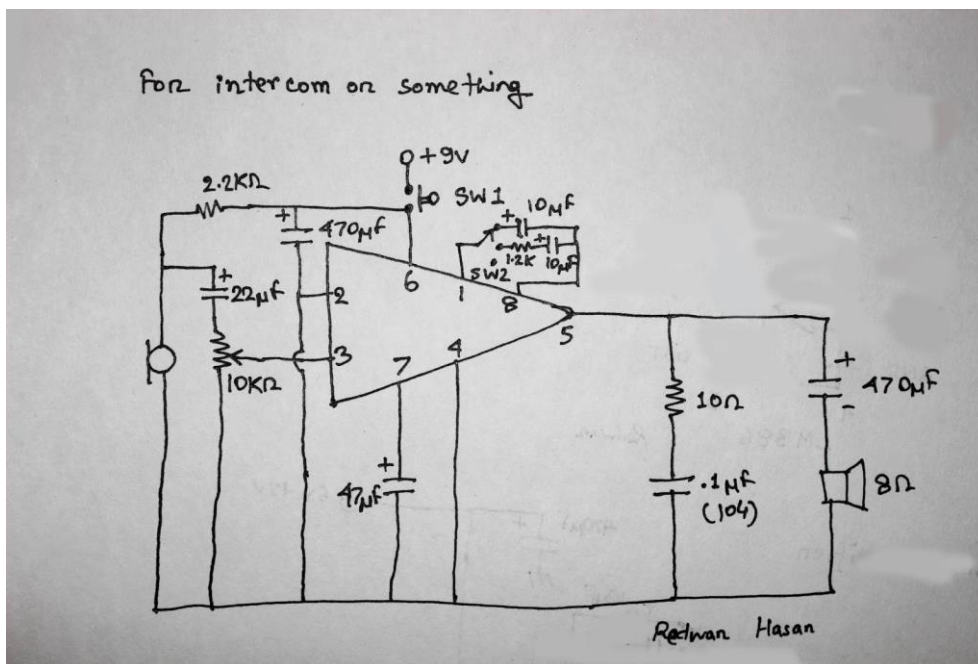
- a. The schematic diagram shall not use four wire junctions. This screams amateur.
- b. The schematic diagram shall use well known units of measurement including:
 - Amps for electrical current - abbreviated capital A
 - Farad for capacitance - abbreviated capital F
 - Henry for Inductance - abbreviate capital H
 - Ohm for resistance - abbreviated with the Greek letter capital omega Ω
 - Watt for power - abbreviated capital W
- c. The schematic diagram should use acceptable SI prefixes when denoting units of measure.
 - Example *55pF* for 55 pico-Farads.
- d. The schematic diagram shall not capitalize any SI Prefix unless there is a 0% chance it could be mistaken by technical and non-technical staff alike.
 - Reason: SI prefixes are case sensitive.
 - Example *55pF* is 55 pico-Farads ($\times 10^{-12}$), but *55PF* is 55 peta-Farads ($\times 10^{15}$)... a very different number.

- It is not acceptable that pico-Farads is the obvious value to an electrical engineer, electrical technician or electrical assembler;
 - Those who purchase parts and handle administrative tasks will take you literally.
- e. The schematic diagram should lead decimal values with a zero and a period.
- Good Example: *0.1uF*
 - Good Example: *0.047uF*
 - Bad Example: *.1uF*
 - Bad Example: *.047uF*
- f. The schematic diagram shall not contain a space between the numerical value and the SI unit
- "28k" is preferred to "28 k"
 - Reason: It is easier to parse values in various post processing needs when the values and their scaling units are combined with no "white space" between them.
 - Note when writing papers and other documents that are simply written words, you should adhere to good English practices and separate the numerical value and the scaling unit with a single space.
 - The description in your Bill of Material shall follow the same rules for English writing and place a single space between the numerical value and the SI scaling value.
 - The scaling value shall be combined with the Unit.
- g. Reference Designators shall be composed two strings: an alpha prefix and a numerical suffix. The prefix shall be one or more uppercase letters, A-Z. The suffix shall by an integer number.
- Reason: Keeping this simple is paramount to allowing your assembler to easily know where you want which part.
 - There have been some reports of schematic packages adding long strings to parts which are common to several repeated sheets within a schematic. Altium Designer 6 is one example. This behavior shall be regarded as a schematic capture program defect and should not be acceptable to proper schematic documentation. The fact that OrCAD Capture and its DOS ancestors have been handling complex hierarchy schematics for decades with no unusual issues suggests modern software design packages are attempting to subvert well understood practices to accommodate mal-designed software. Buyer beware.
 - Example: C3 - Capacitor #3

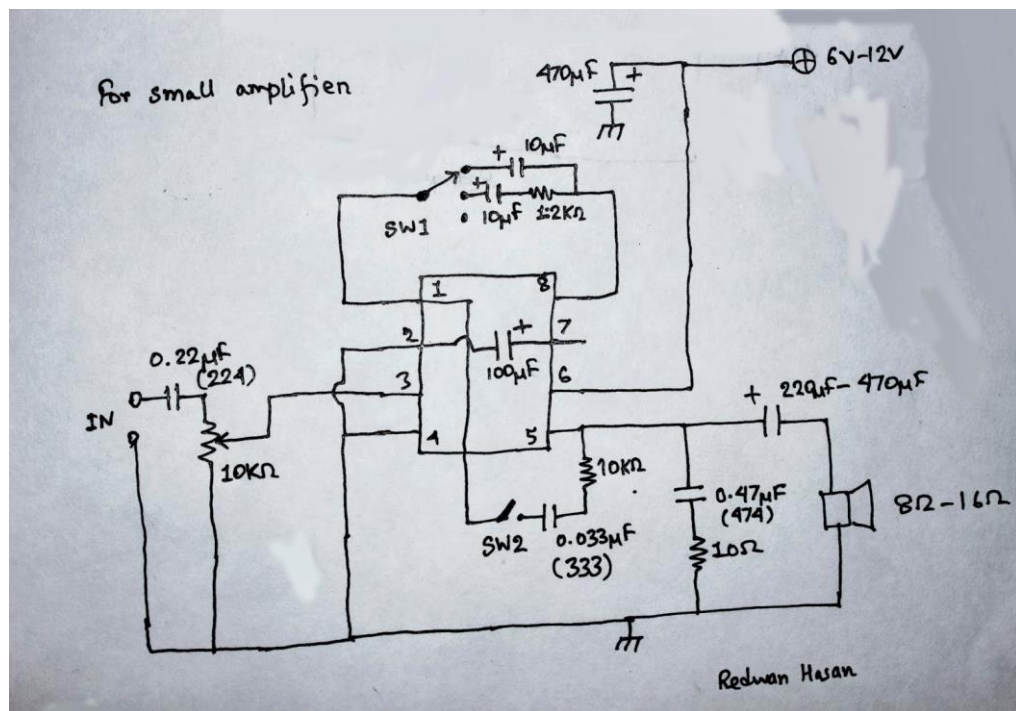
- Example: R156 - Resistor #156
- Example: TP45 - Test Point #45
- Chart of common prefixes:
 - B - Batteries and Cells
 - C - Capacitors
 - D - Diodes, Zener Diodes
 - I - Inductors
 - J - Connectors
 - P - Connectors (P is not usually used on PC Boards)
 - Q - Discrete Transistors, BJT, FET, MOSFET, etc.
 - VR - (NOT PREFERRED) Potentiometers and other variable resistance devices.
 - R - Resistors, Potentiometers, etc.
 - TP - Test Points
 - U - Integrated Circuits

2.4 Exercise: Produce schematic diagrams from the following rough sketches

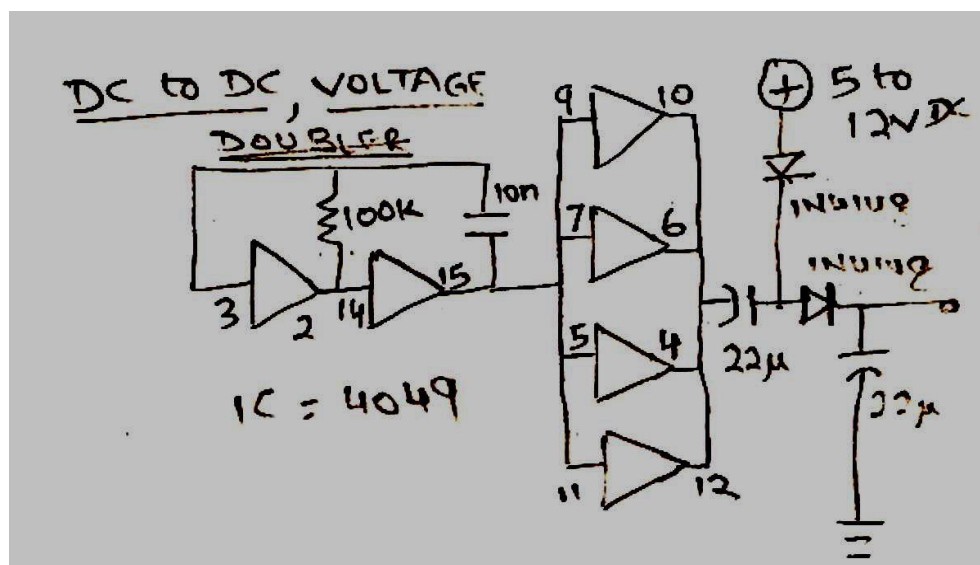
i.



ii.



iii.

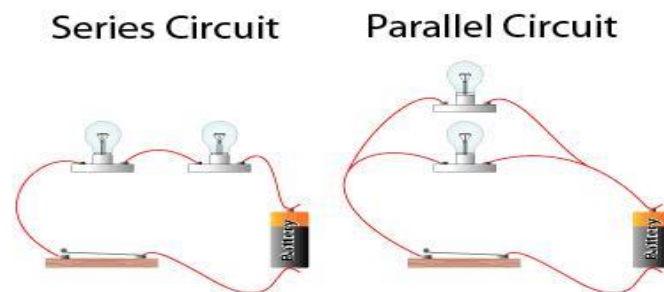


2.5. Component Connection Diagram

Component connection diagram known as pictorial diagram is one in which the components so interconnected are represented using abstract, graphic drawings or realistic pictures. Interconnecting conductors run from one component to the other.

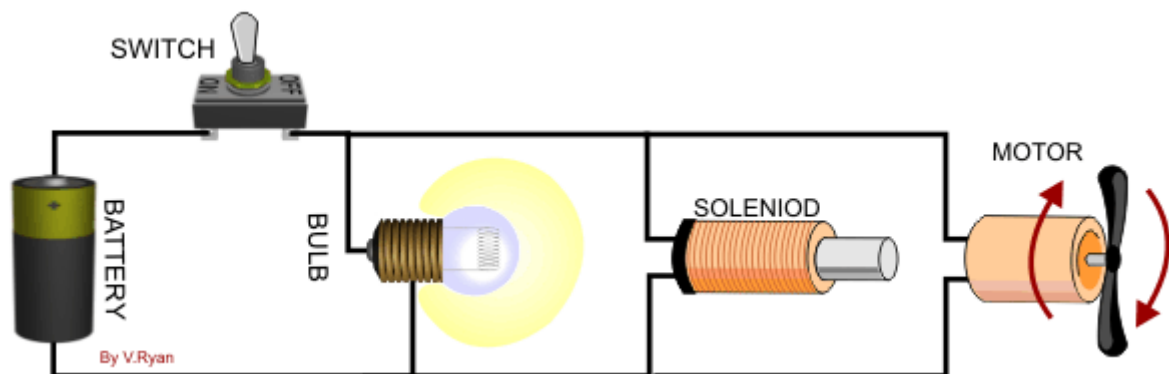
2.51. Examples of Component Diagrams:

i.

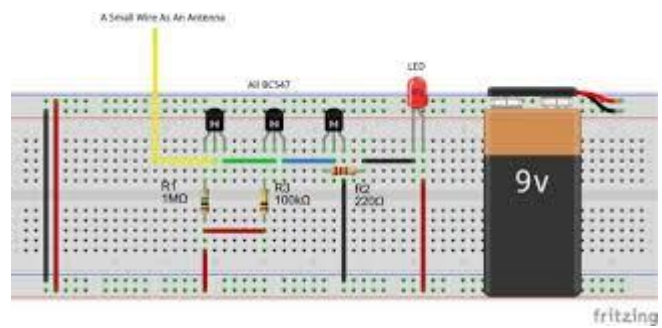


ii.

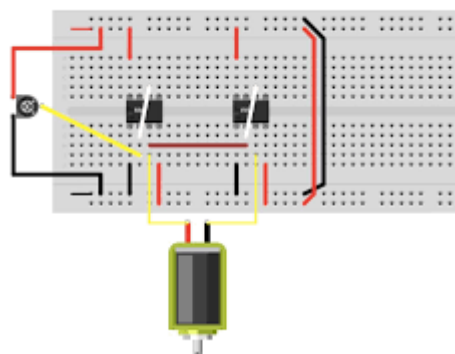
iii.



iv.



v.



vi.



2.6. Part List Tabulation

For proper record and as an input to production a part list of a circuit is generated, usually tabulated giving component details. Details may include name, type, label, value, quantity and comment. A typical component list for the circuit below is given in table 2 sec2.6

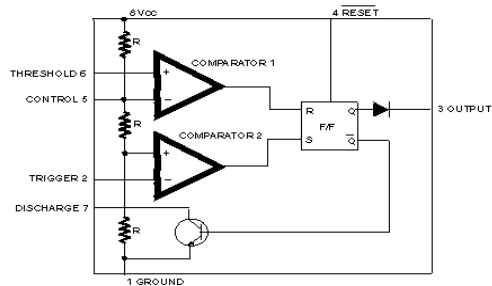
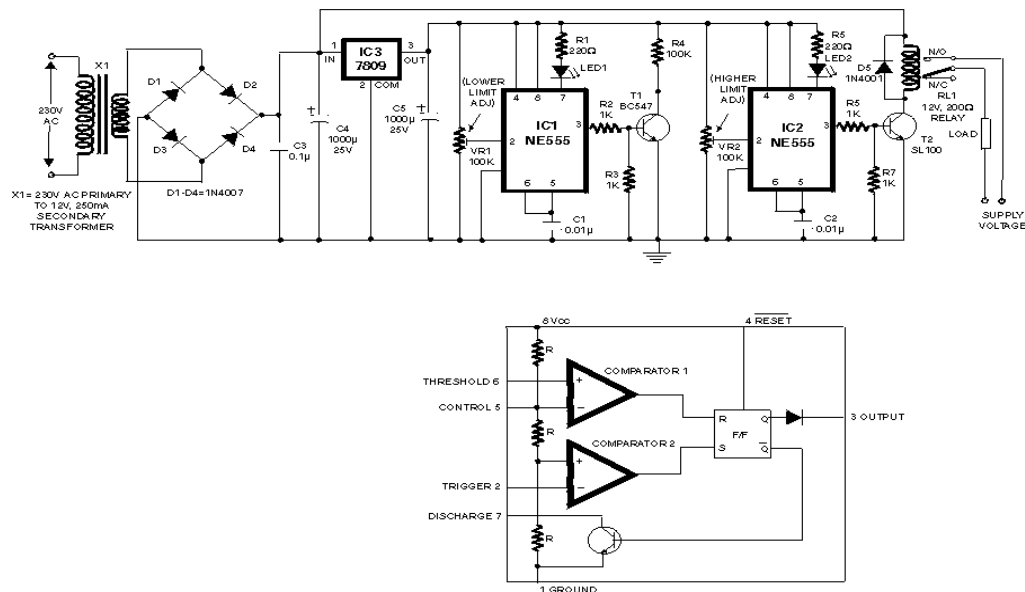


Table sec2.6 Component List

S/N	Component Name	Component Description	Label	Quantity
1	Step down transformer	220v/12, 500mA	X1	1
2	Rectifier diode	1N4007	D1-D4	4
3	Ceramic capacitor	0.1uF	C3	1
4	Electrolytic capacitor	1000uF, 25V	C4,C5	2
5	Fixed Voltage Regulator	LM7809	IC3	1
6	Variable Resistor	100K	VR1,VR2	2
7	Timer IC	LM555	IC1,IC2	2
8	Ceramic capacitor	0.01uF	C1,C2	2
9	Fixed Resistor	220	R1,R5	2
10	Light emitting diode	Red LED	LED1,LED2	2
11	Fixed resistor	1K	R2,R3,R5,R7	4

12	Transistor	BC547	T1,T2	2
13	Rectifier Diode	1N4001	D5	1
	Relay	9V relay 200	RL1	1

The component table can be extended to include and estimate of the cost of the components, in which case the table becomes a bill of engineering measurement and evaluation. BOM can define products as they are designed (engineering bill of materials), as they are ordered (sales bill of materials), as they are built (manufacturing bill of materials), or as they are maintained (service bill of materials).

Table sec2.6 BEME

S/N	Component Name	Component Description	Label	Quantity	Unit cost(#)	Amount(#)
1	Step down transformer	220v/12, 500mA	X1	1	500	500
2	Rectifier diode	1N4007	D1-D4	4	200	800
3	Ceramic capacitor	0.1uF	C3	1	150	150
4	Electrolytic capacitor	1000uF, 25V	C4,C5	2	200	400
5	Fixed Voltage Regulator	LM7809	IC3	1	300	300
6	Variable Resistor	100K	VR1,VR2	2	100	200
7	Timer IC	LM555	IC1,IC2	2	400	800
8	Ceramic capacitor	0.01uF	C1,C2	2	800	160
9	Fixed Resistor	220	R1,R5	2	100	200
10	Light emitting diode	Red LED	LED1,LED2	2	100	200
11	Fixed resistor	1K	R2,R3,R5,R7	4	100	400
12	Transistor	BC547	T1,T2	2	350	700
13	Rectifier Diode	1N4001	D5	1	200	200

	Relay	9V relay 200	RL1	1	500	500
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Total = #5500 (Five thousand five hundred naira only)

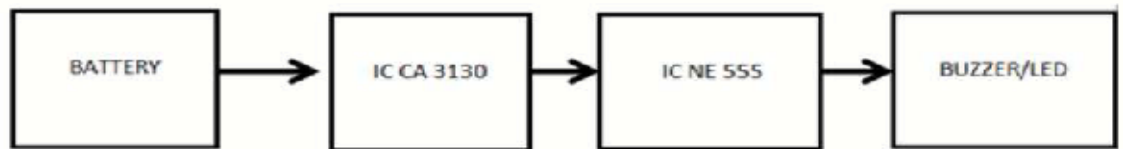
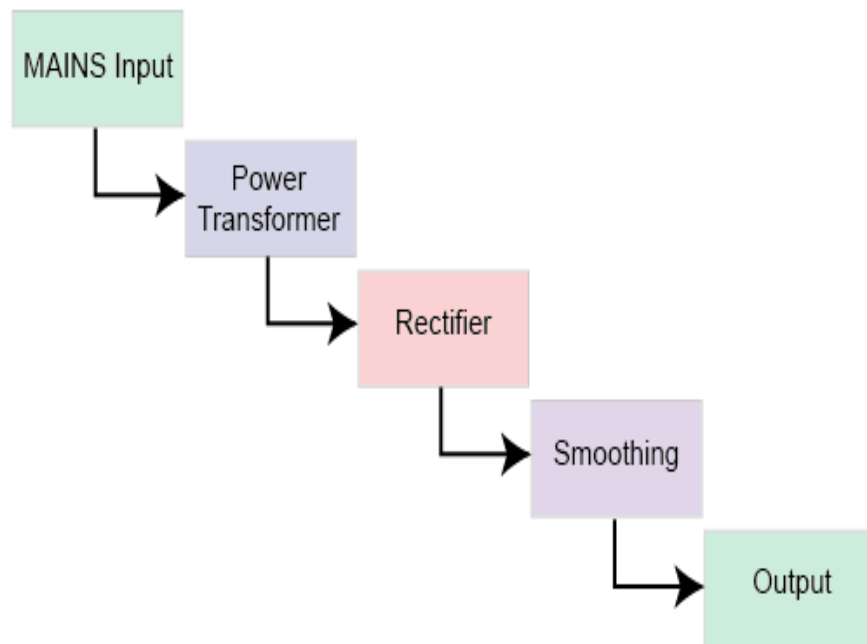
3.0 Block and Logic Diagram

3.1 Block diagram

3.1 A block diagram is a diagram of a system in which the principal parts or functions are represented by blocks connected by lines that show the relationships of the blocks.^[1] They are heavily used in engineering in hardware design, electronic design, software design, and process flow diagrams.

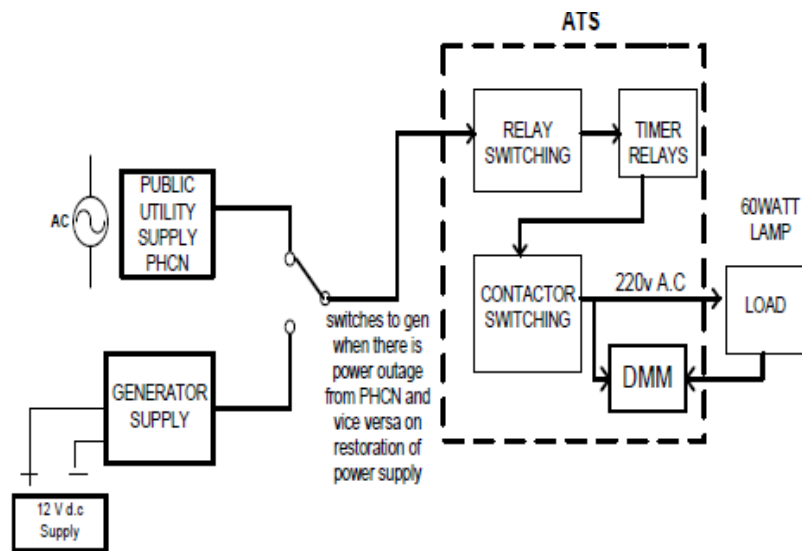
Block diagrams are typically used for higher level, less detailed descriptions that are intended to clarify overall concepts without concern for the details of implementation. Contrast this with the schematic diagrams and layout diagrams used in electrical engineering, which show the implementation details of electrical components and physical construction.

i.



ii.

iii.

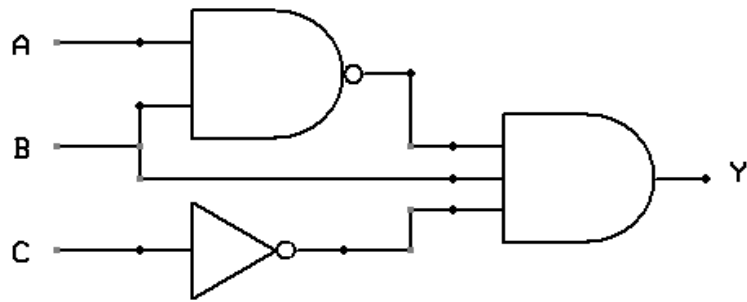


3.2 Logic Diagrams

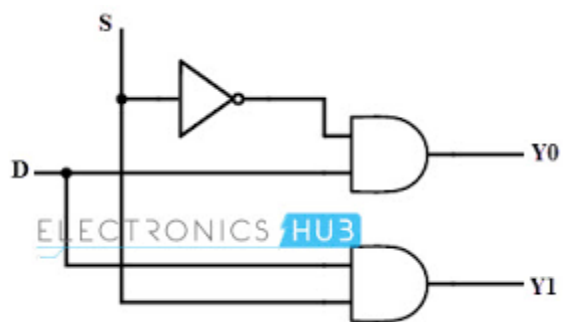
The logic diagram consists of gates and symbols that can directly replace an expression in Boolean arithmetic. A logic gate is a device that can perform one or all of the Boolean logic operations AND, NAND, NOR, NOT, OR, XNOR, and XOR. All types of logic gate, except NOT, accept two binary digits as input, and produce one binary digit as output. NOT gates accept only one input digit.

Each one has a different shape to show its particular function. The inputs (Boolean variables) enter at the left of the symbol and the output leaves from the right. When combined, several gates can make a complex logical evaluation system that has many inputs and outputs. Digital computers are designed by connecting thousands, or millions, of these gates to together.

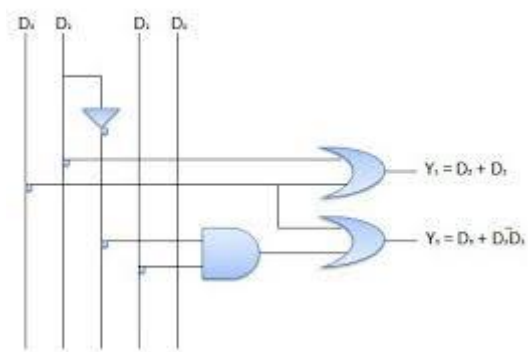
i.



ii.



iii.



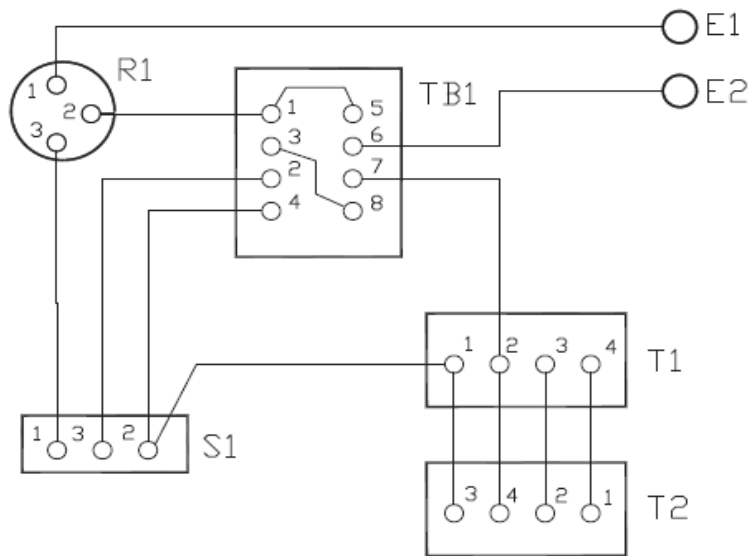
3.3 Procedure for Producing Logic and Block Diagrams

1. Understand the functional assembly.
2. Identify the various stages.
3. Sturdy the signal flow in the block diagram, it should be from left to right and also from top to bottom.
4. The input and output must be shown or must be determined.
5. All the sub connections from feedback loop must be clearly identified and indicated.
6. The engineer or designer usually begins the design of an electronic device by laying out a block diagram.

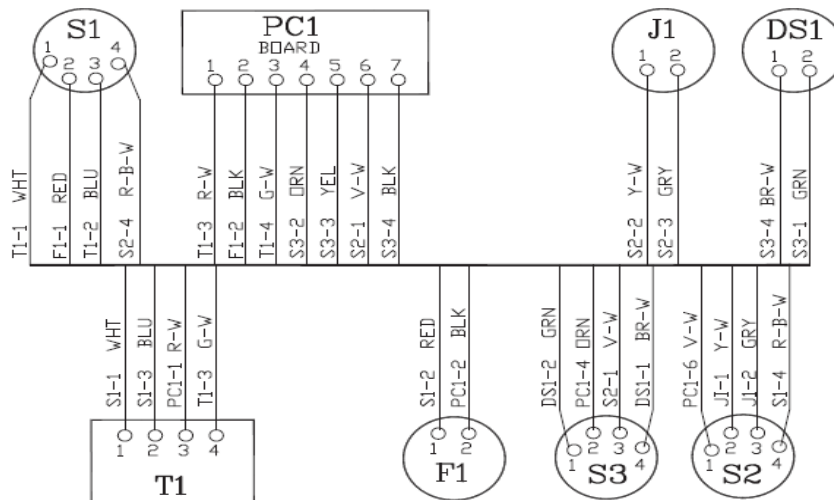
3.4 Types of Wiring Diagrams

A point-to-point diagram is used to show the engineering, manufacturing, and service personnel the wiring between and across components. Point-to-point diagrams contain the information necessary to make or follow all wire connections. This wiring diagram can be shown on the assembly drawing. The diagram on the assembly will be included only if it is practical, and if room is available. The point-to-point drawing will not have a parts list. All the necessary items will be called out on the assembly document. Some point-to-point diagrams show wiring paths on a background of components which are not drawn to scale. The components are drawn out of scale to fit the requirements of a very complex wiring diagram. The wiring is easier to follow when wires are on equal spacing. A second drawing method is to show the components drawn to the true scale.

Point-to-point diagrams show the general physical arrangement of the component parts.

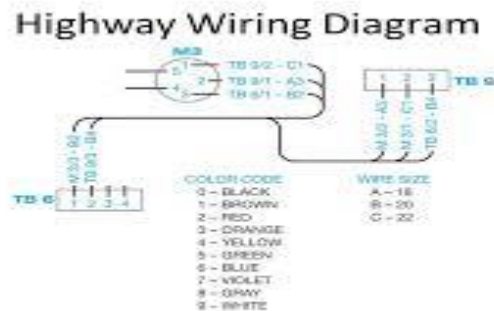


Baseline diagrams are like highway diagrams in that they can handle many wires in an organized manner and they bundle the wires together in one main line. Baseline Wiring diagram drawings are used mostly for service manuals and maintenance books. They are especially good for this kind of information because they can neatly show many lines on a book-size sheet of paper. These drawings will not normally be used for assembly work because the information is too limited.



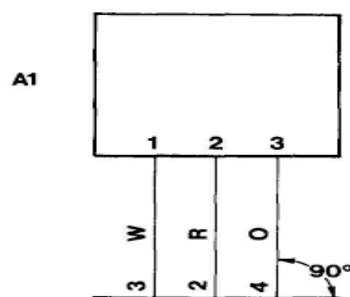
3.5 Difference Between Highway and Bade Line Wiring diagrams

(i) The placement of components- The highway diagram is very concerned with physical placement of the components. The baseline diagram just lines them up in a straight line.



Sketch showing placement and direction of components

(ii) Another difference is in the way the wires enter the main bundle-The highway diagram shows which way the wire will be running in the bundle. The baseline shows wires going 90° into the bundle without disclosing direction.



3.6 Manufacturers Specifications

Manufacturer's specification gives the user the functional limits and inherent parameters of a device like input/output impedance, temperature tolerances, capacitance, inductance etc. Specifications are given to enable effective use of the component in a circuit. The specification provides clear

instructions on project intent, performance and construction. It can reference the quality and standards which should be applied. Materials and manufacturers' products can be clearly defined.

3.7 Exercise:

- i. Produce block and logic diagram from the free sketch given

TUTORIALS

1. . Draw clearly the symbols of the following commonly used electronic components;
(i). Zener diode (ii). Phototransistor (iii).Photo resistor (iv). 2 way switch (v). L.E.D (vi). Variable resistor (vii). Buzzer (viii). NAND gate (ix). NOR gate (x). Thyristor. (xi) Field effect transistor (xii) variable capacitor (xiii) Preset resistor (xiv) Preset capacitor (xv) Potentiometer (xvi) Photo diode (xvii) Inductor (xviii) Tunnel diode (xix) Thermistor (xx) Bell
2. Identify the following passive components with their color codes where applicable;
(i). 10M Ω (ii). 2K2 (iii). 223K, (iv). 104C
3. Define, (i). block diagram (ii). logic diagram
4. Explain step-by- step procedure in drafting block and logic diagram.
5. Draft the logic diagram from the sketch given in figure i.
6. State and explain briefly, types of boards commercially available for building your project.
7. List the step-by-step procedure in laying out and building a circuit neatly on a strip board.
8. State three advantages and three disadvantages of printed circuit board over point-to-point or interconnection using wire
9. Briefly describe the following processes of producing a printed circuit board; (i). Print and etch method
(ii). Photo-reduction method (iii). Any other method you know.
10. Use any method known to you to design to any scale the layout of a single stage universal bias transistor amplifier with input, output and emitter bypass capacitor.
11. Outline the guideline for drawing a well-arranged schematic diagram.
12. Produce the schematic diagram from component connection diagram in figure ii.
13. Produce the part list for the diagram in figure ii.

14. Design a system for wireless video distribution for a four-room apartment, using among the following available devices, audio amplifier, video mixer, video camera, audio mixer, video sender, video monitor, FM transmitter, loudspeaker, and microphone.
15. Explain what manufacturers' specification mean for a device.
16. Design a PA system for use where mains supplied voltage is 110V, for a client who requires 500 Watts of audio power at the cheapest rate. Use available units given in Table i.
17. Write short notes on the following; (i). Single-sided type (ii). Double sided type
(iii). Multilayer structure
18. Define each of the following types of diagram; (i). Point-to- point (ii). Baseline (iii). Highway
19. What are the specifications to be considered when choosing? (i). Audio cable (ii). Mixer
20. Define the following terms (i) signal flow (ii) component reference (iii) sub-assemblies
21. What do you understand by the term design
22. What is the meaning of drafting

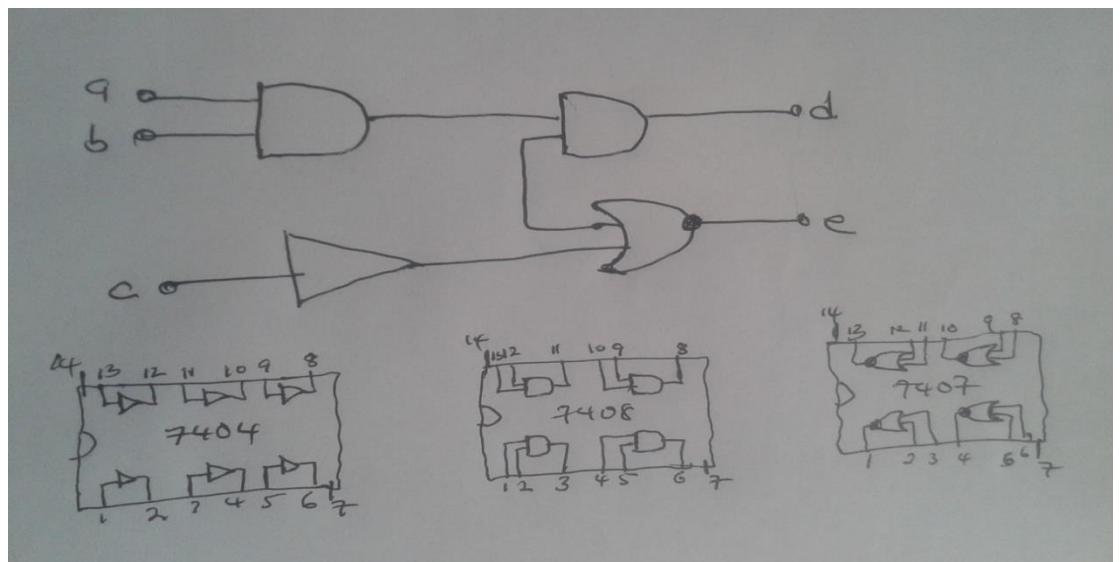


Figure i.

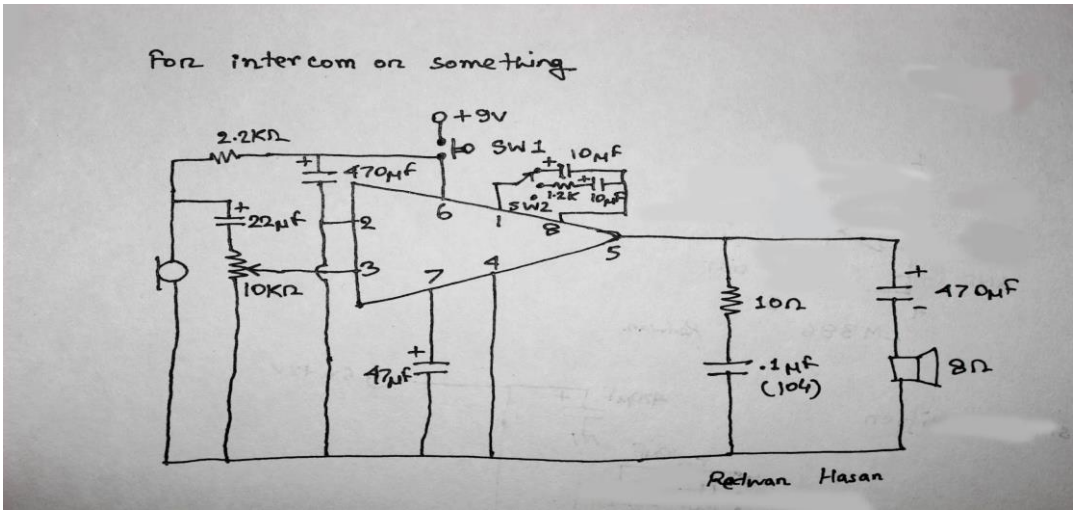


Figure ii

Table i.

Units	A	B	C	D		
Microphone			Impedance-600Ω Cost= #1,200	Impedance-600Ω Cost= #1,300	Impedance-800Ω Cost=#1,000	Impedance-300Ω Cost=#550
Amplifier			P out-500W Input impedance-600Ω Output impedance-8Ω Voltage110v	P out-700W Input impedance-600Ω Output impedance-8Ω Voltage=110v	P out-500W Input impedance-600Ω Output impedance-8Ω Voltage=220v	P out-550W Input impedance-500Ω Out impedance-4Ω Voltage=220v
Speaker			Power-250W Impedance-16Ω	Power-300W Impedance-8Ω Cost=# 2,500	Power-500W Impedance-8Ω Cost=# 5,000	Power-500W Impedance-4Ω Cost=#2,200

	Cost=# 1,000			
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Table ii

Units	A	B	C	E
Microp hone	Impedance-600Ω Cost = #1,200	Impedance-600Ω Cost = #1,300	Impedance-800Ω Cost = #1,000	Impedance-300Ω Cost = #550
Mixer	Input impedance = 600Ω Output Impedance = 600Ω No of inputs = 2 Voltage = 220v Cost = #4,200	Input impedance = 800Ω Output Impedance = 600Ω No of Inputs =3 Voltage=110v Cost = #6,200	Input Impedance =300Ω Output Impedance = 600Ω No of inputs =3 Voltage=110v Cost = #5,200	Input impedance = 800Ω Output Impedance = 600Ω No of inputs =2 Voltage =220v Cost = #5,200
Amplifi er	P out-500W Input impedance- 600Ω Output impedance- 8Ω Voltage110v Cost = #5,000	P out-700W Input impedance- 600Ω Output impedance- 8Ω Voltage=110v Cost = #6,200	P out-500W Input impedance-600Ω Output impedance-8Ω Voltage = 220v Cost = #5,900	P out-550W Input impedance-500Ω Out impedance-4Ω Voltage = 220v Cost = #6,500
loudspe aker	Power-250W Impedance-16Ω Cost = #1,000	Power-300W Impedance-8Ω Cost = #2,500	Power-500W Impedance-8Ω Cost = #5,000	Power-500W Impedance-4Ω Cost = #2,200